

Safety data sheet

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BASF Safety data sheet
Date / Revised: 24.10.2022
Product: **Plurafac® LF 221**

Version: 5.0

(30044056/SDS_GEN_TH/EN)

Date of print: 11.08.2023

1. Substance/preparation and manufacturer/supplier identification

Product name:
Plurafac® LF 221

Use: Raw material for the chemical-technical industry

Manufacturer/supplier:

BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Care Chemicals
Telephone: +49 621 60-57579
E-mail address: emd-ems-ehs-masterdata@basf.com

Emergency information:

International emergency number:
Telephone: +49 180 2273-112

2. Hazard identification

Classification according to UN GHS 2009

Classification of the substance and mixture:

Acute toxicity: Cat.4 (oral)
Skin corrosion/irritation: Cat.3
Serious eye damage/eye irritation: Cat.2A
Hazardous to the aquatic environment - acute: Cat.2
Hazardous to the aquatic environment - chronic: Cat.3

Label elements and precautionary statement:

Pictogram:

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Signal Word:
 Warning

Hazard Statement:

H319	Causes serious eye irritation.
H316	Causes mild skin irritation.
H302	Harmful if swallowed.
H412	Harmful to aquatic life with long lasting effects.
H401	Toxic to aquatic life.

Precautionary Statements (Prevention):

P280	Wear eye protection.
P273	Avoid release to the environment.
P270	Do not eat, drink or smoke when using this product.
P264	Wash contaminated body parts thoroughly after handling.

Precautionary Statements (Response):

P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P301 + P312	IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell.
P330	Rinse mouth
P337 + P313	If eye irritation persists: Get medical attention.
P332 + P313	If skin irritation occurs: Get medical attention.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.
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Other hazards which do not result in classification:

No specific dangers known, if the regulations/notes for storage and handling are considered.

This surfactant complies with the biodegradability criteria as laid down in Regulation (EC) No.648/2004 on detergents. Data to support this assertion are held at the disposal of the competent authorities of the Member States and will be made available to them at their direct request or at the request of a detergent manufacturer.

3. Composition/information on ingredients

Chemical nature

Substance nature: Substance

| Alcohols, C13-15-branched and linear, butoxylated, ethoxylated
 | CAS Number: 111905-53-4

4. First-Aid Measures

General advice:

Remove contaminated clothing.

If inhaled:

Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:

Wash thoroughly with soap and water

On contact with eyes:

Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:

Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Note to physician:

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Suitable extinguishing media:

water spray, dry powder, foam

Specific hazards:

harmful vapours, carbon oxides

Evolution of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

Special protective equipment:

Wear a self-contained breathing apparatus.

Further information:

The degree of risk is governed by the burning substance and the fire conditions. Contaminated extinguishing water must be disposed of in accordance with official regulations.

6. Accidental Release Measures

Personal precautions:

Use personal protective clothing. Information regarding personal protective measures, see section 8.

Environmental precautions:

Contain contaminated water/firefighting water. Do not discharge into drains/surface waters/groundwater.

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Methods for cleaning up or taking up:
For large amounts: Dike spillage. Pump off product.
For residues: Pick up with suitable absorbent material.
Dispose of absorbed material in accordance with regulations.

Additional information: High risk of slipping due to leakage/spillage of product.

7. Handling and Storage

Handling

Shut containers immediately after taking product because product takes up the humidity of air.

Protection against fire and explosion:
No special precautions necessary.

Storage

Suitable materials for containers: High density polyethylene (HDPE), Low density polyethylene (LDPE), Stainless steel 1.4301 (V2), Stainless steel 1.4306 (V2A), Stainless steel 1.4361, Stainless steel 1.4401, Stainless steel 1.4541, Stainless steel 1.4571, Stove-lacquer RDL 50, Stainless steel 1.4439, Stainless steel 1.4539

Further information on storage conditions: Keep container tightly closed and dry; store in a cool place.

Storage stability:
Storage temperature: $\leq 70\text{ }^{\circ}\text{C}$

The packed product is not damaged by low temperatures or by frost. Bulk must be protected from solidification.

Protect from temperatures above: $70\text{ }^{\circ}\text{C}$

Properties of the product change irreversibly on exceeding the limit temperature.

8. Exposure controls and personal protection

Components with occupational exposure limits

| No substance specific occupational exposure limits known.

Personal protective equipment

Respiratory protection:

Respiratory protection in case of vapour/aerosol release. Particle filter with medium efficiency for solid and liquid particles (e.g. EN 143 or 149, Type P2 or FFP2)

Hand protection:

Chemical resistant protective gloves

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Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):
 nitrile rubber (NBR) - 0.4 mm coating thickness

Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.
 Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:

Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:

Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:

Wearing of closed work clothing is required additionally to the stated personal protection equipment.
 No eating, drinking, smoking or tobacco use at the place of work. Handle in accordance with good industrial hygiene and safety practice.

9. Physical and Chemical Properties

Form:	liquid	
Colour:	colourless to yellowish	
Odour:	alcohol-like	
Odour threshold:	not determined	
pH value:	approx. 7 (50 g/l, 20 °C)	(DIN EN 1262)
solidification temperature:	approx. 5 °C	(DIN ISO 2207)
Boiling point:	> 250 °C	(estimated)
Flash point:	> 200 °C	(ISO 2592)
Evaporation rate:	Value can be approximated from Henry's Law Constant or vapor pressure.	
Flammability (solid/gas):	not self-igniting	
Lower explosion limit:	For liquids not relevant for classification and labelling., The lower explosion point may be 5 - 15 °C below the flash point.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	
Ignition temperature:	> 300 °C	
Thermal decomposition:	> 350 °C	(DTA)

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Self ignition:	not self-igniting	
Self heating ability:	It is not a substance capable of spontaneous heating according to UN transport regulations class 4.2.	
Explosion hazard:	not explosive	
Fire promoting properties:	not fire-propagating	
Radioactivity:		not radioactive for transport purposes
Vapour pressure:	< 0.1 hPa (20 °C)	(BASF method)
Density:	approx. 1.00 g/cm ³ (20 °C)	(DIN 51757)
	approx. 0.97 g/cm ³ (70 °C)	(DIN 51757)
Relative density:	No data available.	
Relative vapour density (air):	not determined	
Solubility in water:	soluble	
Miscibility with water:	miscible in all proportions	
Hygroscopy:	hygroscopic	
Solubility (qualitative) solvent(s):	distilled water, Ethanol, isopropanol, petroleum spirit	
	soluble	
Partitioning coefficient n-octanol/water (log Pow):	not applicable	
Surface tension:	approx. 30 mN/m (23 °C; 1 g/l)	(DIN EN 14370)
Viscosity, dynamic:	100 mPa.s (23 °C)	(DIN EN 12092)
	approx. 5,000 mPa.s (10 °C)	(DIN EN 12092)

Other Information:

If necessary, information on other physical and chemical parameters is indicated in this section.

10. Stability and Reactivity

Conditions to avoid:

Avoid humidity. See SDS section 7 - Handling and storage.

Thermal decomposition: > 350 °C (DTA)

Substances to avoid:

caustics, halogens, Alkalines, acids, reactive chemicals

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Hazardous reactions:

No hazardous reactions when stored and handled according to instructions.

Hazardous decomposition products:

No hazardous decomposition products if stored and handled as prescribed/indicated.

Chemical stability:

The product is stable if stored and handled as prescribed/indicated.

11. Toxicological Information

Routes of exposure

Acute oral toxicity

Experimental/calculated data:

LD50rat (oral): > 300 - 2,000 mg/kg (similar to OECD guideline 401)

Acute inhalation toxicity

LC50 rat (by inhalation):

not determined

Acute dermal toxicity

LD50 rat (dermal):

not determined

Assessment of acute toxicity

Of moderate toxicity after single ingestion.

Symptoms

Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.

(Further) symptoms and / or effects are not known so far

Irritation

Assessment of irritating effects:

Skin contact causes slight irritation. Eye contact causes irritation.

Experimental/calculated data:

Skin corrosion/irritation rabbit: Slightly irritating. (similar to OECD guideline 404)

Serious eye damage/irritation rabbit: Irritant. (similar to OECD guideline 405)

Respiratory/Skin sensitization

Assessment of sensitization:

No data available.

Germ cell mutagenicity

Assessment of mutagenicity:

The substance was not mutagenic in bacteria. The substance was not mutagenic in a test with mammals.

Carcinogenicity

Assessment of carcinogenicity:
Not classified, due to lack of data.

Reproductive toxicity

Assessment of reproduction toxicity:
Not classified, due to lack of data.

Developmental toxicity

Assessment of teratogenicity:
Not classified, due to lack of data.

Specific target organ toxicity (single exposure)

Remarks: No data available.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
No adverse effects were observed after repeated exposure in animal studies.

Aspiration hazard

No aspiration hazard expected.

12. Ecological Information

Ecotoxicity

Assessment of aquatic toxicity:
Toxic to aquatic life. Harmful to aquatic life with long lasting effects.

Toxicity to fish:
LC50 (96 h) > 1 - 10 mg/l, *Leuciscus idus* (static)

Aquatic invertebrates:
EC50 (48 h) > 1 - 10 mg/l, *Daphnia magna* (semistatic)

Aquatic plants:
EC50 (72 h), algae
not determined

Microorganisms/Effect on activated sludge:
EC10 > 1,000 mg/l, activated sludge (DEV-L2)

Chronic toxicity to fish:
No data available.

Chronic toxicity to aquatic invertebrates:

No observed effect concentration (21.0 d), > 0.1 - 1 mg/l, Daphnia magna (OECD Guideline 202, part 2, semistatic)

Assessment of terrestrial toxicity:

No data available concerning terrestrial toxicity.

Mobility

Assessment transport between environmental compartments:

The substance will not evaporate into the atmosphere from the water surface.

Adsorption to solid soil phase is possible.

Persistence and degradability

Assessment biodegradation and elimination (H₂O):

Readily biodegradable (according to OECD criteria).

Readily biodegradable (according to OECD criteria).

Elimination information:

> 70 % DOC reduction (28 d) (aerobic, activated sludge) Readily biodegradable.

> 60 % CO₂ formation relative to the theoretical value (28 d) (aerobic, activated sludge) Readily biodegradable.

Bioaccumulation potential

Bioaccumulation potential:

Accumulation in organisms is not to be expected.

Other adverse effects

Adsorbable organically-bound halogen (AOX):

This product contains no organically-bound halogen.

Additional information

Add. remarks environm. fate & pathway:

Treatment in biological waste water treatment plants has to be performed according to local and administrative regulations.

Other ecotoxicological advice:

Inhibition of degradation activity in activated sludge is not to be anticipated during correct introduction of low concentrations. Do not release untreated into natural waters.

13. Disposal Considerations

Must be disposed of or incinerated in accordance with local regulations.

Contaminated packaging:

Uncontaminated packaging can be re-used.

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Packs that cannot be cleaned should be disposed of in the same manner as the contents.

14. Transport Information

Domestic transport:

	Not classified as a dangerous good under transport regulations
UN number or ID number	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number or ID number	Not applicable
Proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

15. Regulatory Information

Other regulations

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

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16. Other Information

Information on intended use: This product is of industrial quality and unless otherwise specified or agreed intended exclusively for industrial use. This includes the mentioned and recommended usage. Any other intended applications should be discussed with the manufacturer. In particular this concerns the application for products that are the object of special standards and regulations.

Vertical lines in the left hand margin indicate an amendment from the previous version.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

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